Retail Sales Analysis

SQL-Based Business Analysis and Interactive Excel Dashboard

Project Report | Analysis period: 2011–2014

1. Executive Summary

This project analyzes a multi-year retail sales dataset to understand overall business performance, sales growth, product and customer performance, regional contribution, and the relationship between discounting and profitability. PostgreSQL was used to query and validate the data, while Microsoft Excel was used to build an interactive dashboard for exploring key performance indicators and trends.

The dataset contains 51,290 records. Across the full period, the business generated $12.64M in sales and $1.47M in profit, producing an overall profit margin of 11.61%. Sales grew every year from 2011 to 2014. The analysis also highlights a major profitability concern: transactions with discounts above 30% recorded substantial combined losses.

2. Project Objectives

- Measure overall retail performance using sales, profit, margin, orders, customers and quantity.
- Track annual and monthly sales trends and identify growth patterns.
- Compare profitability across markets, categories, customer segments and shipping modes.
- Identify high- and low-performing products and customers.
- Evaluate how discount levels relate to sales and profitability.
- Present the findings in an interactive Excel dashboard.

3. Dataset and Tools

Dataset	Global Superstore-style retail sales dataset
Records analyzed	51,290 rows
Columns	24 fields
Period covered	1 January 2011 – 31 December 2014
Database	PostgreSQL
Visualization	Microsoft Excel PivotTables, PivotCharts and slicers

The dataset includes order and shipping dates, customer and geographic details, product hierarchy, sales, quantity, discount, profit, shipping cost and order priority.

4. Data Quality Assessment

Before analysis, SQL checks were used to validate the loaded row count, uniqueness, missing values, date relationships and numeric ranges.

Check	Result	Interpretation
Total rows loaded	51,290	Matches expected dataset row count
Duplicate Row IDs	0	No repeated Row ID values detected

Missing Order IDs / Order Dates	0 / 0	Key order fields complete
Missing Product IDs	0	Product identifier complete
Missing Postal Codes	41,296	High missingness; retain as unavailable rather than inventing values
Ship date before order date	0	No invalid shipping sequence detected
Negative sales / invalid quantity	0 / 0	No such records detected
Discount outside 0–100%	0	No unusual discount values detected
Negative shipping cost	0	No negative shipping costs detected

Data-quality conclusion: the core order, product and date checks did not reveal missing key identifiers, duplicate Row IDs or invalid date sequences. Postal Code is the notable completeness issue, with 41,296 missing values. This field should be handled carefully in any detailed geographic analysis.

5. Key Performance Indicators

KPI	Overall result
Total sales	$12,642,507.25
Total profit	$1,467,456.55
Profit margin	11.61%
Quantity sold	178,312

Profit margin is calculated as total profit divided by total sales, multiplied by 100.

6. Sales Performance Over Time

Sales increased in every year of the analysis period. Both sales and order counts rose from 2011 through 2014, indicating sustained expansion in transaction volume and revenue.

Year	Sales	Orders	YoY sales growth
2011	$2,259,451.64	4,440	—
2012	$2,677,439.91	5,343	18.50%
2013	$3,405,748.03	6,721	27.20%
2014	$4,299,867.67	8,531	26.25%

2013 recorded the highest year-over-year sales growth at 27.20%. In 2014, sales reached approximately $4.30M, the highest annual total in the dataset.

7. Category and Market Performance

7.1 Category performance

Category	Sales	Profit margin
Technology	$4,744,558.76	13.99%
Furniture	$4,110,875.32	6.94%
Office Supplies	$3,787,073.17	13.69%

Technology led category sales and recorded the highest category margin among the three major categories. Furniture generated substantial revenue but had the lowest margin, making it a priority for deeper review.

7.2 Market profitability

Market	Profit margin
Canada	26.62%
EU	12.69%
US	12.47%
APAC	12.16%
Africa	11.34%
LATAM	10.24%
EMEA	5.45%

Market margins vary considerably. Canada has the highest reported margin, while EMEA has the lowest. A market-level action plan should examine product mix, discounting, shipping costs and local pricing.

8. Discount and Profitability Analysis

Discount group	Sales / volume indicator	Profitability result
No discount	$6,992,410.95	25.32%
Above 30% discount	$10,361 records	-$793,529.65 profit; -51.27% margin

The no-discount group generated approximately $6.99M in sales at a 25.32% margin. In contrast, records with discounts above 30% produced a combined loss of about $793.5K and a -51.27% margin. This comparison supports reviewing high-discount transactions, although it should be treated as an association rather than proof that discounting alone caused each loss.

9. SQL Analysis Performed

- Table creation and data loading structure for the 24-column retail dataset.
- Data quality checks: row counts, unique Row IDs, duplicate records, missing fields, date validity and suspicious numeric values.
- Overall KPIs and year-wise sales, profit, order counts and year-over-year sales growth.
- Market, category, sub-category and regional performance.
- Discount-level performance and profit margin comparison.
- Monthly sales trends and strongest months.
- Customer segment and shipping-mode performance.
- Most and least profitable products and customers.
- Country-level sales and profit contribution.

The SQL script is organized around table creation, data-quality validation, overall KPIs, time trends, market and product analysis, discount analysis, customer analysis and country-level performance.

10. Excel Dashboard

The Excel dashboard brings the main measures and comparisons into one interactive view. It uses KPI cards, PivotCharts and slicers to let a viewer filter the analysis by category, year and month.

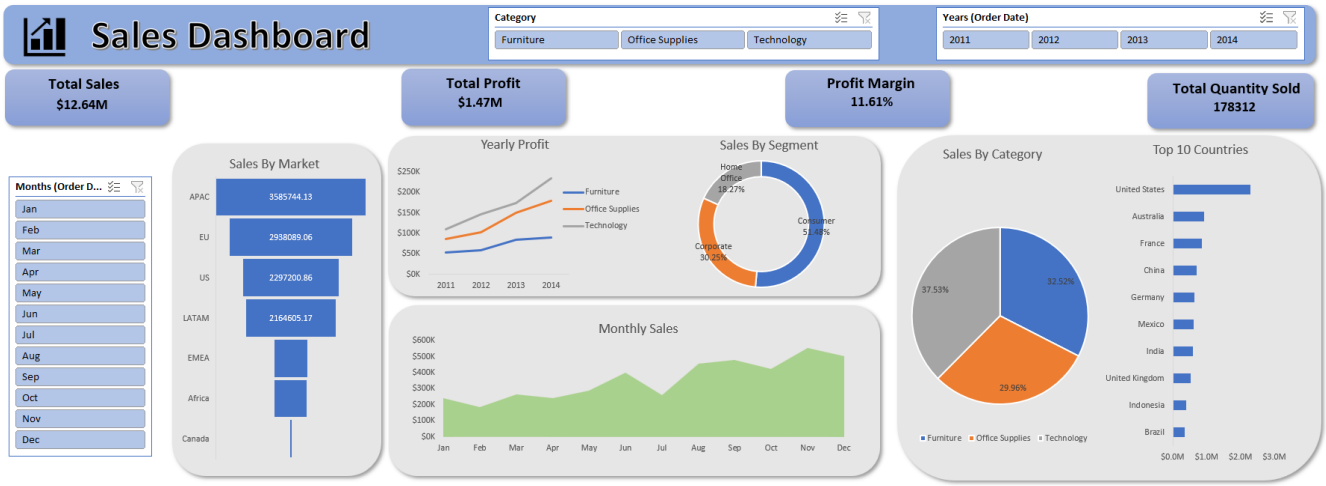


Figure 1. Retail Sales Analysis interactive Excel dashboard.

- KPI cards for Total Sales, Total Profit, Profit Margin and Total Quantity Sold.
- Category and year slicers for interactive filtering.
- Month slicer to explore monthly performance.
- Sales by Market horizontal bar chart.
- Yearly Profit by Category line chart.
- Sales by Segment doughnut chart.
- Sales by Category pie chart.
- Top 10 Countries bar chart.
- Monthly Sales area chart.

11. Key Findings

1. Consistent sales growth: Annual sales increased from $2.26M in 2011 to $4.30M in 2014. Sales growth was positive in each year, with the strongest year-over-year increase in 2013 at 27.20%.

2. Technology leads category sales and margin: Technology generated approximately $4.74M in sales and had a 13.99% profit margin. Office Supplies also delivered a healthy 13.69% margin.

3. Furniture has a profitability gap: Furniture generated approximately $4.11M in sales but recorded a lower 6.94% profit margin than the other two categories. It deserves closer review at the sub-category and product level.

4. Heavy discounting is associated with losses: Records with discounts above 30% generated a combined loss of approximately $793.5K and a negative 51.27% margin. This is a strong signal to review high-discount transactions and approval rules.

5. Market profitability varies: Canada showed the highest reported market margin at 26.62%, while EMEA had the lowest at 5.45%. Sales volume alone should therefore not be used to judge market performance.

6. Missing postal codes need careful treatment: Postal Code is missing in 41,296 records. This can limit postal-level mapping and location analysis. Missing values should not be filled with guessed postal codes.

12. Business Recommendations

1. Review discounts above 30% by product, customer, market and order size. Introduce approval thresholds or minimum-margin checks before granting exceptional discounts.
2. Investigate Furniture sub-categories and the least profitable products to identify whether pricing, discounting, shipping cost or product mix is reducing margins.

3. Study the practices behind stronger-margin markets such as Canada, and investigate the lower-margin performance in EMEA before making expansion or pricing decisions.
4. Protect profitable Technology and Office Supplies product lines while monitoring product-level profitability rather than relying only on category sales.
5. Improve geographic data completeness at source. Keep missing postal codes clearly marked as missing and exclude them from postal-level analysis where necessary.
6. Continue monitoring sales, profit, margin, orders and quantity together. Revenue growth should be evaluated alongside profit and margin.

13. Limitations and Considerations

- The report summarizes the analysis results supplied for the project. Product-level and customer-level result tables are queried in SQL but their full ranked outputs were not provided for this report.
- Postal Code is missing for a large share of records, so postal-code-level geographic conclusions are not appropriate without additional source data.
- The dashboard screenshot is a point-in-time view. Slicer selections can change the displayed values; the headline project KPIs in this report represent the overall dataset unless stated otherwise.
- The analysis is descriptive. It identifies associations, such as high discounts occurring alongside losses, but does not by itself prove that discounting caused every loss.

14. Conclusion

This project demonstrates an end-to-end retail sales analysis workflow using PostgreSQL and Excel. The SQL analysis validates the dataset and evaluates performance across time, categories, markets, discount levels, products and customers. The Excel dashboard makes the results easier to explore. The most important business opportunities are to protect margins when offering discounts, investigate Furniture profitability and address weaker market margins. Overall, the project shows how structured data analysis can turn transactional records into practical business insights.